

Color-wall: Adaptive Color Filter to Reduce Digital Distractions during PC Work

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ABSTRACT

As the Attention Economy has grown, numerous digital contents are made with the intention of capturing users' attention and spending more of their time by the use of attractive graphics and color psychology. In this paper, we propose a novel defense strategy, called a *color-wall*, to block digital distraction brought on by the consumption of visually seductive and addictive digital content. When it detects that a user is viewing media or using applications unrelated to the main task, it removes colors from the screen (gray-scaling) to make them less aesthetically pleasing and directs the user away from looking at them. We implemented a mac OS software as a proof-of-concept for the *color-wall* and conducted user testing (N=14) to assess the effectiveness of the proposed strategy. The evaluation results suggest that the *color-wall* is an effective strategy for reducing unwanted digital distractions such as browsing restricted content (YouTube, Twitter, etc.) while working on a PC.

CCS CONCEPTS

• Human-centered computing → Human computer interaction (HCI); Ubiquitous and mobile computing.

KEYWORDS

digital distractions, digital addiction, color filter, disengagement, friction nudges, making un-use

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1 INTRODUCTION

The COVID-19 pandemic has integrated digital technology into our daily lives at an unprecedented rate. With the promotion of digital education, computer-based learning is now the norm for

many students [19]. This paradigm shift has also increased our concerns about our digital well-being [4]. For example, digital distraction, which is the frequent use of digital content such as social media for off-task purposes, and chronic multitasking overload are representative concerns that lead to a reduced attention span and motivation to learn [6, 15, 21]. In fact, resolving these concerns is not easy, as many social media companies, including YouTube, Twitter, and Facebook, design addictive content to maintain users' ongoing attention [11, 14]. To practice healthy digital well-being, it is essential to consider how to better deal with digital content designed to exploit psychological vulnerabilities to capture attention and time.

In this paper, we propose a novel defense strategy, called a *color-wall*, that prevents unintended digital distractions by adaptively removing the color of visually seductive and addictive digital contents. Our idea is inspired by firewall design [13] to protect against unauthorized access from the outside and the concept of making something unusable so that the user disengages with the technology [10]. As a first step in evaluating the effectiveness of the color wall, we implemented a Proof-of-Concept (PoC) model as mac OS software, focusing on the everyday computer work of college students. Our PoC model of the *color-wall* is designed according to the MAPE-K (Monitor, Analyze, Plan, Execute, and Knowledge) control loop [12], the software consists of four mechanisms where (1) *monitor* content used in the workspace, (2) *analyze* whether digital distractions occur, (3) *plan* change of color filter, and (4) *execute* color filter control, where digital content that poses a threat to users is predefined as *knowledge*. We evaluated the effectiveness of the proposed approach through user studies (N=14). The evaluation results suggest that the *color-wall* is an effective strategy for reducing unwanted digital distractions such as browsing restricted content (YouTube, Twitter, etc.) while working on a PC.

2 RELATED WORK

Over the past few years, research on behavioral change support systems that use nudges [20] has been increasing in the area of ubiquitous computing [1, 3, 17]. Several strategy have also been proposed for digital detox. For example, existing research employs strategies such as locking the app [7] or continuously vibrating [18] when users exceed voluntary time limits for using social media. To discourage the use of the app, a strategy is proposed to insert a lockout task (e.g., typing a fixed length number) with a near-zero workload each time the app is launched [8]. Also, a self-monitoring system has been proposed for the purpose of improving individual productivity by grasping the computer usage status and feeding back the results [9]. However, existing explicit approaches based

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on restrictions, alerts, and feedback carry the risk of reactance, including creating counterproductive effects.

Following the concept of making un-use proposed by Alina et al.[10], we employ aesthetic transformations that decrease the content use experience by undermining the visual appeal of the media or application itself. Specifically, we take advantage of the fact that many media and companies carefully design interfaces based on color psychology to attract the user's attention and adaptively remove colors from the screen (gray-scaling). In fact, setting the screen of the device on a grayscale has emerged as a promising way to discourage the use of smartphones, apps and media [2, 5]. However, the process of activating the grayscale varies from device to device, and switching manually is a bit of a hassle, so it will likely not become a habit for many people, except those who are highly motivated to detox digitally. In addition, in the case of PC work, there are many cases where full color is desired for some applications, such as program development, so constantly using a grayscale screen is an unacceptable option from the standpoint of keeping of concentration and work efficiency. Therefore, we believe it is desirable to adaptively switch the use of color filters according to the working situation to increase user acceptability.

3 ATTENTION-WALL

3.1 Use Case Scenario

We provide realistic use-case scenarios showing how *color-wall* can be useful in practice, as shown in Figure 1. Taro is a college student who tends to rely on digital content (Figure 1A). Often, he spends hours browsing social media while trying to make progress with his programming assignments (Figure 1B). One day, when he told a friend about his digital distraction problem, he was told about the *color-wall*. Upon hearing about it, Taro decided to give it a try (Figure 1C). At the start of the work, he sets in advance a restricted list of apps that may trigger digital distractions and begins to use them. After an hour of work, when his concentration was waning, he accidentally reached for YouTube, as is his usual habit. At this time, the *color-wall* application detects the digital distraction operation on the PC and turns on the color filter. With the screen now in grayscale, Taro remembers his original purpose and begins working on his programming assignments again (Figure 1D). After several days of using the app, Taro is gradually becoming able to practice digital well-being (Figure 1E). His entertainment-related screen time is gradually decreasing, and he is happy that he is able to detox digitally (Figure 1F).

3.2 Design and Implementation

As a first step in evaluating the effectiveness of the color wall, we implemented a Proof-of-Concept (PoC) model as mac OS software, focusing on the everyday computer work of college students. Herein, we aim to nudge people to reduce digital distractions during PC work through the aesthetic transformations that turn color filters on or off depending on the content the user uses on the computer workspace. Our idea is inspired by firewall design [13] to protect against unauthorized access from the outside and the concept of making un-use [10] to create the unused that disengages with technology.

As shown in Figure 2, *color-wall* is designed according to the MAPE-K (Monitor, Analyze, Plan, Execute, and Knowledge) control loop [12], the software consists of four mechanisms where (1) *monitor* content used in the workspace, (2) *analyze* whether digital distractions occur, (3) *plan* change of color filter, and (4) *execute* color filter control, where digital content that poses a threat to users is predefined as *knowledge*. When the user is using a work-related application, the *color-wall* keeps the color filter OFF (Figure 2 phase 1). If the user starts to use a restricted application or media that is not related to his/her work, the *color-wall* detects the action and changes the state of the color filter from OFF to ON (Figure 2 phase 2). The user's screen will then be in grayscale with the colors removed (Figure 2 phase 3). Subsequently, if the user no longer uses the restricted application, *color-wall* will change the state of the color filter from ON to OFF (Figure 2 phase 4). Here, the system returns to the phase1 state.

The *color-wall* software was implemented by Swift. The acquisition of information about the active window on the PC was implemented using the Quartz Window Services API ¹. Control of color filters, such as turning grayscale on and off on the screen, was done via NSAppleScript ². The blacklist, which indicates apps to be restricted, is maintained in a CSV format file, and users can add any app or URL to the list through the app's UI. This allows users to customize which apps to restrict (grayscale is enabled when that app is opened) based on their work.

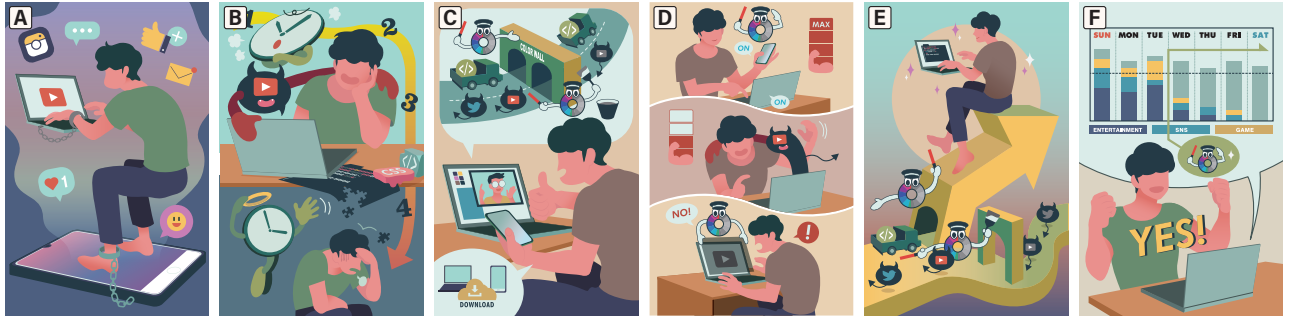
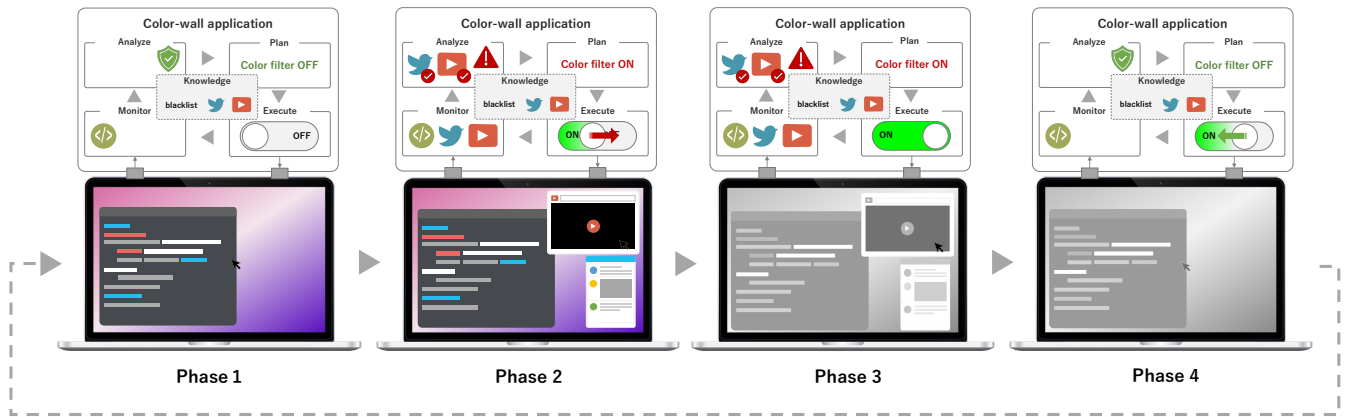
4 EXPERIMENT

4.1 Settings

This user study intended to gain insight into how feedback from *color-wall* is being experienced. To achieve this goal, we conducted a within-subjects design experiment in which participants installed the *color-wall* application on their own PCs and worked freely for 7 hours (between 10:00 - 19:00) on different days with and without automatic grayscale. Specifically, we set two conditions: the proposed app records only work logs (NF: no filter condition), and the proposed app records work logs and performs adaptive color filter control (CF: color filter condition). We investigated the changes in the log of apps and browsers used under each condition and how the experience of the grayscale mode being automatically turned on when a digital distraction occurs would change the user's awareness. The former was measured from the *color-wall* application logs and the latter from the post-experiment questionnaire. 14 students (age: Mean = 22.71, SD = 1.13) in the same university participated in this experiment. All participants have never used this application before and will be using it for the first time in this experiment. We divided the participants into two groups to counterbalance the two conditions, NF and CF. Group A was then assigned conditions in the order CF, NF and Group B in the order NF, CF. Each participant in the experiment was asked in a pre-experiment questionnaire to indicate the media (application names and website names) that could induce digital distraction during academic work and to launch the *color-wall* application with these media added to a restricted list. In

¹Quartz Window Services: https://developer.apple.com/documentation/coregraphics/quartz_window_services

²NSAppleScript: <https://developer.apple.com/documentation/foundation/nsapplescript>

Figure 1: The use case scenario of *color-wall*.Figure 2: Overview and working example of *color-wall* software.

other words, under CF conditions, participants will experience an aesthetic transformation in which grayscale mode is automatically turned on by the *color-wall* app when they use media associated with digital distractions. For the purpose of comparing concentration and productivity for each condition, participants were also asked to record their subjective concentration and productivity (on a scale of 1 to 10) every hour for the duration of the experiment.

4.2 Results and Discussion

A summary of the experimental results is shown in Figure 3. Here we show the total time spent using the restricted (a) and unrestricted apps (b) in each condition, as well as the subjective concentration (c) and productivity (d) results answered by the users at hourly intervals. Results show a trend toward decreasing usage time per application and a slight increase in concentration and productivity. In particular, the average usage time of restricted apps was 0.35 hours (SD = 0.69) for the NF condition and 0.13 hours (SD = 0.35) for the CF condition. From this result, it was confirmed that the restricted app usage time was decreased by 13.26 minutes (- 63.74%) under the CF condition compared to the baseline under the NF condition. To analyze significant differences in the NF and CF conditions of each result, we performed a paired-samples two-sided Wilcoxon signed rank test [22]. The test results showed a significant

difference in the usage time of the restricted apps ($p = 0.044$, $p < 0.05$). On the other hand, no significant differences were identified for the other results. These results suggest that the feedback of the color wall application may act as a trigger to reduce the use of irrelevant restricted applications such as YouTube and Twitter while working on a PC. The lack of negative changes in subjective concentration and productivity suggests that the strategy of *color-wall* is acceptable and does not interfere with the main task.

The application usage logs indicated that the participants were acting to avoid grayscale screens. In other words, it means that as soon as the *color-wall* turned on the color filter, they stopped using restricted applications that were not related to the main task to avoid it. The main tasks that the participants were engaged in during the experiment were coding programs and making the presentation material related to their research activities. Since these tasks have aspects of using color to improve work efficiency (syntax highlighting function in the program editor, etc.), we consider that they tended to abandon their use of the restricted apps to maintain the color of their workspace. In the follow-up questionnaire after the experiment, we got interesting opinions and impressions about the *color-wall* application. Many participants enjoyed the new experience and we received considerable positive feedback such as “I didn’t feel like watching the YouTube videos that I always watch

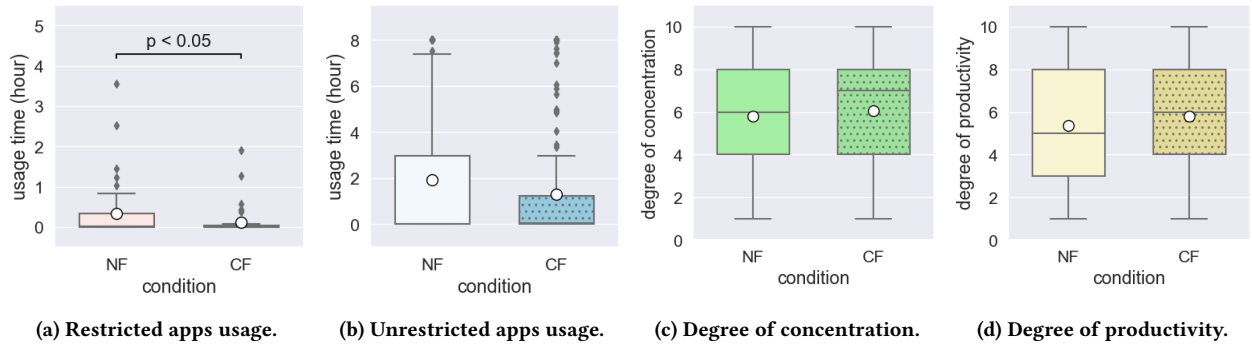


Figure 3: Summary of experimental results.

when it comes to grayscale. Because of this, I felt like I was working more than usual.”, “I find myself spending much less time on Twitter with the gray filter on, and I feel more focused.”, “Overall, I spend less time distracted by things other than work.”, “It helped me realize that I was losing focus.”, and “I was not exposed to more irrelevant information than necessary. I had less interest in the screen and less use of the PC during breaks.” When asked when they would like to use *color-wall*, respondents indicated that they would like to use *color-wall* when they need to catch up on their work, or when they are on deadline for an assignment or paper. These results support that the strategy of *color-wall* is likely helpful in reducing digital distractions not related to the main work. On the other hand, some participants revealed their experiences, “I usually enjoy YouTube content by ear as background music, so I didn’t feel anything when it was grayscaled”, and “I heard a notification sound and reflexively opened a social networking site.” These indicate the need for an approach that utilizes not only visual information but also auditory information.

5 CONCLUSION AND FUTURE WORK

This paper proposes *color-wall* as a new resistance approach to prevent digital distractions caused by the use of visually seductive and addictive digital content. Specifically, when it detects that a user is viewing media or applications that are not relevant to the main task, it will nudge in the direction of un-use them by removing colors from the screen (making them grayscale) and reducing their visual appeal. We implemented a Proof-of-Concept of the *color-wall* as a mac OS app and evaluated the effectiveness of the proposed approach through user studies (N=14). The evaluation results suggest that the *color-wall* is an effective tool for reducing unwanted digital distractions such as browsing restricted content (Youtube, Twitter, etc.) while working on a PC. In the future, we will extend the implementation of *color-wall* to cover multi-device environments, including mobile devices such as smartphones used by users. Finding ways to enhance people’s digital well-being in a multi-device environment remains a challenging research topic [16]. We also plan to investigate the effects of habituation and boredom through long-term, in-the-wild experiments with a larger number of participants.

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